البرمجة المتقدمة | اسئلة السنوات السابقة مع حلها

انقر للإنتقال

اسئلة المد 2024 صباحي

اسئلة المد 2024 مسائي

اسئلة الفاينال 2023 الدول الأول

اسئلة الفاينال 2023 الدول الثاني

اسئلة متفرقة

Write java program to create avstudent(st[],d[][]) to find the sum and av for the following arryas of students

Ali 10, 10,20
Moh 20,100,50
Hassan 100,100,100

```
public class ok{
    public static void main(String[] args){

        String [] students = {"Ali", "Moh", "Hassan"};
        int de[][] = { {10,10,20}, {20, 100, 50},
                       { 100, 100, 100} } ;
        avstudent (students, de) ;
    }

    static void avstudent (String[] arr , int x[][]
    {   int sum = ;
        for (int i=0 ; i<3; i++){
            sum =0;
            for (int j=0; j<3 ; j++)
                sum += x [i] [j] ;

            System.out.println("name="+ arr[i]
                                + "sum="+sum
                                + "av="+ (sum/3));
        }
    }

}
```

Q3: Write java program to create array x[]={20,30,5,2,1} have integer value

1- Pass x[] array to method **Searchkey(x, key)** to find key=5; then print index of it.

2- pass x[] array to method **Sortar**() to array in Ascending order.

3- search key =5 before sort array and after sorting.

Q3:

```
public class Q3{
    public static void main(String[] args){
     int x[] = {20, 30, 5, 2, 1};
     SearchKey(x, 5);
     Sorter(x);
     SearchKey(x, 5);
    }
    static void searchKey(int[] arr, int Key){
        for(int i = 0; i < arr.length; i++){
            if (arr[i] == Key)
                System.out.print("found at index "+i);
        }
    }
    static void Sortar(int [] arr){
        int temp = 0;
        for(int i=0; i< arr.length; i++)
            for(int j=i+1; i< arr.length; j++){
                if (arr[i] > arr[j]){
                    temp = arr[i];
                    arr[i] = arr[j];
                    arr[j] = temp;
                }
            }
        }
}
```

Second Exam

Q1: - Write java program to create array Price [4].

1- input from keyboard any prices

2- make discount 50% for each price in array then save results in new array discount[] then return new array and print result in main method.

Q1:

```
import java.util.Scanner;
public class Q1{
    public static void main(String[] args){
        Scanner in = new Scanner(System.in);
        double price[] = new double[4];

        System.out.print("write prices :");
        for (int i=0 ; i<4 ; i++)
            price[i] = in.nextDouble();

        for(int i=0 ; i<4 ; i++)
            System.out.println(discount(price)[i]);

    }
    static double[] discount(double[] arr){
        double[] temp= new double [4];
        for (int i=0 ; i<4 ; i++)
            temp[i]= arr[i]/2 ;
        return temp;
    }

}
```

اسئلة المد صباحي 2024 مع حلها

Q1:- Write (True) or (False & Correct it) **(2.5 Marks)**

1- All the items of array have different datatype.

2- When call method should be has same number of parameters and different data type.

3- The indexes of the array start with zero always

4- The lengths of the rows in 2D array are required to be the same.

5- Sqrt is a method of Scanner class.

****

Q2: Write java program to create array of 10 Names then passed this array to **(3 Marks)**

1- Find (Names [], Key) to find key("Ali") and print results in method.

2- Sort array using SortName(Names[]) then print sorted array

3- Use (**Switch**) to make options for methods

****

Q3: Create a class called Invoice, this class have **(2.5 Marks)**

1- **Create** array **items**[] ={ sugar , Tea, rice , oil}

2- Create array called **Information**

1	200
2	400
3	600

3- **Find Maximum and minimum**

Q4:- Correct the following of code . (2 Marks)

```
Int m ;

Int x[  ]=int [10];

Int [ . ] y = int [11] ;

For(m=1; m<=10; ++ m )

X[m]=y[m]=m;

x=y=new int [20];
x[3]=y[3]
for ( m=1 ; m<=10; ++m )

System . out .Println ( x) ;
                    x [m]
```

Q1/

1- false same datatype

2- false diffrent or same data type

3- true

4- False , aren't required to be same

5- false , math class

Q2/

```
import java.util.Scanner;
public class mid1 {
  static void Find(String Names[], String Key) {
    int flag = 0;
    for(int i = 0 ; i < Names.length ; i++) {
        if (Names[i] == Key)
            flag = 1;
    } // for
  if (flag == 1)
      System.out.println("found");

  else
      System.out.println("not found");
  } //method
```

تابع للسؤال الثاني ...

```
static void SortName(String[] names) {
    String temp;

    for (int i=0 ; i< names.length ; i++)
        for(int j=i+1 ; i<names.length ; j++) -
        {   if ( names[j] < names[i] ){
                temp = names[i] ;
                names[i] = names[j] ;
                names[j] = temp ;

            } //if
        } //for

    for (int i=0 ; i< names.length; i++)
        System.out.println( names[i]);
} //method
```

```
                String Names[] = { ....., .....};
public static void main (String[] args){
    Scanner in = new Scanner (System.in) ;
    System.out.println ("1. search Ali , 2. Sort  your choice : ");
    int choice = in.nextInt() ;
    switch (choice){
        case 1 :
            Find( Names, "Ali") ; break ;
        case 2 :
            SortName(Names); break ; } //switch
}//main
}//public class
```

Q3

```
public class Invoice {
public static void main (String args) {
  String[] items = {"sugar", "Tea", "rice", "oil"};
  int[][] Information = { {1, 200},
                          {2, 400}, {3, 600}};
  int minimum = Information[0][1];
  int Maximum = Information[0][1];
  for (int i = 0 ; i < 3 ; i++) {
     if (Information[i][1] < minimum)
        minimum = Information[i][1];
     if (Information[i][1] > Maximum)
        Maximum = Information[i][1];
  }//for

  System.out.println(minimum);
  System.out.println(Maximum);

}//main

}//public class
```

Q4/

```
int m;
int x[] = new int[10];
int [] y = new int[10];
for ( m=0 ; m < 10 ; ++m)
      x[m] = y[m] = m;

x = y = new int [20];
for ( m=0 ; m < 20 ; ++m)
      System.out.println(x);
```

x[m]

اسئلة المد المسائي 2024 مع حلها

Q1:- Write (True) or (False & Correct it) **(2.5 Marks)**

1- All the items of array have same datatype. F No one data type

2- Static methods can access to static variables. F can't

3- The indexes of the array start with zero always T

4- The size of 1D array can determine using Size ArrayName F length

5- Power method is a method of Scanner class. F for power number

****

Q2: H.W:-1 Write Java Program to input numbers then create Method sqrtNumber(A) and powerNumber(a,b)

Use (Switch) to select of methods (1: SqrtNumber 2: PowerNumber).

****

Q3: Write java program to create avstudent(st[],d[][]) to find the av for each student for the following arrays of students Names and grades. **(2.5 Marks)**

Mohamed	**10**	**10**	**20**
Ali	**20**	**100**	**50**
Jaafar	**100**	**100**	**100**

Q4:- Correct the following of code .

```
Int m ;
Int x[ ]=int [10];
Int [ ] y = int [11] ;
For(m=1; m<=10; ++ m )
X[m]=y[m]=m;
x=y=new int [20];
for ( m=1 ; m<=10; ++m )
System . out .Println ( x) ;
```

new int

new int

y

0

10

20

x[m]

x[m]

Q1/

1- true

2- true

3- true

4- false, ArrayName.length

5- false, ~~pow method~~ math

Q2/

```
import java.util.Scanner;
public class m {
  static void sqrtNumber (int A){
    float result = math.sqrt(A);
    System.out.println(result); } // method sqrt

  static void powerNumber (int A, int B){
    ~~for (int i=1; i<=B; i++)~~
      ~~A=~~

    int result = math.pow(A, B);
    System.out.println(result); } //power method

  public static void main (String[] args) {
    Scanner in = new Scanner (System.in);
    System.out.println("1 for sqrt  2 for power  write choice : ");
    int choice = in.nextInt();
  switch(choice){
      case 1:
          int X = in.nextInt();
          sqrtNumber(X); break;

      case 2:
          int x = in.nextInt();
          int x2 = in.nextInt();
          powerNumber(X, x2); break; } //switch
  } //main
} //public class
```

Q3/ public class Q3 {

```
    static void avstudent(String st[], int d[][]) {
float av ;
    For (int i=0; i < d.length ; i++) {
        For (int j=0 ; j <d.length ; j++)
        {
            av = av + d[i][j] ;
        }
        av = av / 3 ;
        String name = st[i] ;
        System.out.println( name + " av is " + av);
        av =0 ;
    } // for
} // method
public static void main (String[] args) {
String st[] = { "Mohamed", "Ali", "Jaafar" } ;
int x[][] = { {10,10,20}, {20, 100,50}
            , {100, 100, 100} } ;
avstudent (st, x) ;
} // main
} // public class
```

Q4/

```
int m;
int x[] = new int[10];
int [ ] y = new int[10];
for (m=0 ; m<10 ; ++m)
    x[m] = y[m] = m;

x = y = new int [20];

for ( m=0 ; m < 20 ; ++m)
    System.out.println(x[m]);
```

اسئلة الفاينال 2023 الدور الثاني مع حلها

University of Kufa

Faculty of Computer Science & Math's

Department: COMPUTER SCIENCE

Course Name: ADVANCE JAVA

Academic Year: 2022-2023

Course Stage/Level: 1

Date of Exam: / /2023

Time: TWO HOURS

Final Exam

NOTE: ANSWER FOUR QUESTIONS ONLY (THIRD QUESTION COMPULSORY)

Q1: Write java program to create method spilt(S1) to split string in two parts. (10 M)

1- Pass string from main method S1= "**I would like to eat KFC**"

2- **return results s1 ,s2 to main method**

Q2: Write java program to create array books[][]={{"Java", "200$"}, {"C#", "300$"}, {"Python", "400$"}}; (10 M)

1- find book with price (200$)

2- print his name only after finding it.

Q3: Write java program to create method **Arth(char opp[])** with passing array Op[]={+, -, *, /, ^} then **apply all this mathematic operations on variables (A , B).** (10 M)

Call method in main.

Q4: *write java program to create three methods area of circle, circumference of a circle, and area of area of the triangle, pass parameter not return value.* (10 M)

Q5: Write java program using passing array names[] to get the following output

Output:

ali : **Ali**

mohammed: **Mohammed**

jaafar: **Jaafar**

Ali Jaafar , Mohammed

أ.م. د فرقان حسن صالح
رئيس القسم

أ.م. د علي عبد الكريم حبيب
أستاذ المادة

Q1/

```
public class f2023 {
    static String[] split ( String S1) {
        String[] r = S1.split ("to");
        return r;
    } //method
    public static void main (String[] args) {
        String S1 = "I would like to eat KFC";
        String[] a = split (S1);
        System.out.println( a[0]);
        System. out .println (a[1]);
    } //main

} // public class
```

Q2

```
public class Q2 {
  public static void main(String[] args){
   String[][] books = { {"java", "200$"}
                      , {"C#", "300$"}
                      , {"python", "400$"}} ;
    for (int i=0 ; i<3 ; i++) {
      if ( books[i][1] == "200$")
        System.out.println (books[i][0]) ;
    } //for

  } //main

} //public class
```

Q3/

```
public class Q3{

 static void Arth(char[] oop){
  int A= 10;
  int B= 5;
  for (int i =0 ; i < oop.length ; i++) {
     switch (oop[i]) {
        case '+' :
        System.out.println(A+B); break;

        case '*' :
           System.out.println(A*B); break;

        case '-' :
           System.out.println(A-B); break;

        case '/' :
           System.out.println(A/B); break;

        case '^' :
           System.out.println(A^B); break;
     }//switch
  }//for
 public static void main(String[] args){
 char Op[] = {'+','-','*','/','^'};
  Arth(Op); }//main
}//public class
```

Q4/ import java.util.Scanner;

```
public class Q4{

    static void areaOfCircle(int r){
        float area = 3.14 * r * r ;
        System.out.println(area); } // method

    static void cirOfCircle (int r){
        float cir = 2 * 3.14 * r ;
        System.out.println ( cir); } //method

    static void areaOfT(int b, int h) {
        float area = 0.5 * b * h ;
        System.out.println (area); } //method

    public static void main( String [] args) {
        Scanner in = new Scanner (System.in) ;
        System.out.println("1. area of circle   2. cir of circle
                    3. area of tringle   : ") ;
        int choice = in.nextInt();
        switch (choice){
            case 1:
                int r = in.nextInt(); areaOfCircle (r); break;
            case 2:
                int r = in.nextInt(); cirOfCircle (r); break;
            case 3 : int b = in.nextInt();
            int h = in.nextInt (); areaOfT (b,h); break;
```

Q5)

```
public class Q5 {
  static void ss( String[] names) {
    for (int i=0 ; i<names.length ; i++) {
        String first= names[i].substring (0,1) ;
        String rest = names[i]. substring ( 1 ) ;
        System.out.println( names[i] + " : "+ first.toUpperCase()
                                  +rest ) ;

    } //for

    for (int i=0 ; i<names.length ; i++) {
        String first = names[i] . substring (0, 1) ;
        String rest = names[i]. substring (1) ;
        System.out.print( first.toUpperCase() + rest + "    " ) ;
    } //for
  } //method
public static void main ( String[] args ) {
    String names[] = { "ali", "mohammed", "Jaafar"} ;
    ss (names) ;
  } //main
} //public class
```

اسئلة الفاينال 2023 الدور الأول مع حلها

Q1: Write java program to create array (10 M)

X =										
200	100	300	150	99	170	120	33	40	1	55

Then enter three cases **using switch** (**1:** sum items of array from 170 to end of array.
2: Input any key from keyboard then check it found or not found in array.
3: Sort items from 200 to 170).

```
import java.util.Scanner;
class HelloWorld {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
        int x[] = {200,100,300,150,99,170,120,33,40,1,55};
        System.out.println("1 or 2 or 3 : ");
        int choice = in.nextInt();
        switch(choice){
            case 1:
                int sum = 0;
                for (int i=5 ; i < x.length ; i++)
                    sum+= x[i];
                System.out.println(sum);
                break;
            case 2:
                int fromkeyboard = in.nextInt();
                int flag = 0;
                for (int i=0 ; i< x.length ; i++){
                    if (x[i] == fromkeyboard)
                        flag = 1;
                }
                if (flag == 1)
                    System.out.println("found");
                else
                    System.out.println("not found");
                break;
            case 3:
                int temp ;
                for (int i=0 ; i <6 ; i++)
                    for (int j=i+1 ; j < 6 ; j++){
                        if (x[j] < x[i]){
                            temp = x[i];
                            x[i] = x[j];
                            x[j] = temp;
                        }
                    }
                break;
        } //switch


    } //main
} //public class
```

استخدام دالة

Q2: Write **java program** to get the following **output**. Using **(Prime Method)** **(10 M)**

```
true,  true,  true,  true,  false, true,
true,  true,  true,  true,  false, true,
true,  true,  true,  true,  false, true,
true,  true,  true,  true,  false, true,
false, false, false, false, false, false,
true,  true,  true,  true,  false, true,
```

```java
import java.util.Scanner;
class HelloWorld {
    static boolean isPrime(int number){
        int flag = 0 ;
        for (int i = number-1 ; i>1 ; i--){
            if (number%i == 0){
                flag = 1;
                break;
            }
        }
        if (flag == 1)
            return false;
        else
            return true;
    }
    public static void main(String[] args) {
        int[][] numbers = {{1,2,3,4,5,6},{7,8,9,10,11,12},{13,14,15,16,17,18},
            {19,20,21,22,23,24},{25,26,27,28,29,30},{31,32,33,34,35,36}};
        for (int i =0 ; i < 6 ; i++){
            for (int j =0 ; j<6 ; j++){
                System.out.print(isPrime(numbers[i][j])+",  ");
            }
            System.out.println("");
        }
    } //main
} //public class
```

Q3: Write java program to create method **MerregArray**(int **A1**[],int **A2**[]) and **(10 M)**

create **Sortarray**(int merrgedarray[]) , **let A1[4] , A2[4]**

يرجع مصفوفة واحدة بعد الدمج

1- Return result as one **array** to main method.

2- Sort result using **Sortarray**(int **MerrgedArray**[ ]) . رتبة المصفوفة

```java
class HelloWorld {
    static int[] MerregArray(int A[] , int B[]){
        int NewArray[] = new int[8];
        for(int i =0 ; i<4 ; i++){
            NewArray[i] = A[i];
            NewArray[i+4] = B[i];}
        return NewArray;
    }
    static void Sortarray(int MerrgedArray[]){
        int temp;
        for (int i=0 ; i<MerrgedArray.length ; i++)
            for (int j=i+1 ; j<MerrgedArray.length ; j++){
                if (MerrgedArray[j] < MerrgedArray[i]){
                    temp = MerrgedArray[i];
                    MerrgedArray[i] = MerrgedArray[j];
                    MerrgedArray[j] = temp;
                }
            }
    }
    public static void main(String[] args) {
        int A[] = {1,2,3,4};
        int B[] = {5,6,7,8};
        int[] after = MerregArray(A,B);
        Sortarray(after);
    }
}
```

اختار الاجابة

Q4: Select the right answer **(10 M)**

1- Declaration of array XX has elements of type char

(a- int XX[ ] b- [ ]int XX c- char XX[ ] d- [ ]char XX)

2- All elements of array must be **of the**

(a- same data types b- different data types c- All data types d- integer data types)

3- We can specify the number of elements of the array XX by

(a- XX.size( ) b- XX.size c-XX.lenght( ) d- XX.lenght

4- Let X={ 3,5,6,8,2,8,9,7,11} to find middle of array using

(a-(Length-1)/2 b- length/2 c- length( )/2 d- (Length( )-1)/2)

5- Call method with pass parameter and return value like

(a-M=XX(10,1) b- M=XX( ) c-XX(10,1) d-XX( ))

Q5: Correct the following code

```
A) Public int [] secondarydia (int x [][]){
for (int i=0; i<=3; i++)
    {
        for(int j=0;j<4;++j)
        {  if(i=j)
           System.out.println(x[i][j]);
           if((j-i)==2-1)
        System.out.println(x[i][j] +",,");  }  }}
Public static void main(String args[] ){
int x[]= new {1,2,3,4},{5,6,7,8},{9,10,11,12};
secondarydia( );
```

```
static void seconderydia (int x [][]){
    for (int i=0; i<=2; i++)
    {
        for(int j=0;j<3;++j)
        {  if(i==j)
            System.out.println(x[i][j]);
           if((j+i) == 2)
            System.out.println(x[i][j] + ",,");  }  }}

    public static void main(String args[]) {
        int x[][] = {{1,2,3},{4,5,6},{7,8,9}};
        seconderydia(x);

    }
```

(10 M)

```
B)    class Main {
public static void square(int num) {
return num * num;  }
public static void main(String[] args) {
 int result;
     square( );
System.out.println("Squared value of
10 is: " + result);    } }
```

0,0 0,1 0,2 0,3
1,0 1,1 1,2 1,3
2,0 2,1

```
public class Main {

    public static int square(int num){
        return num*num;}

    public static void main(String[] args) {
        int result = square(10);
        System.out.println("Squared value of 10 is: "+result);} }
```